

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
DATA SCIENCE AND ANALYTICS
(7th SEMESTER, 4TH Year) LAB EXPERIMENT # 4

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Date of Conduct: Submission Date: 8/19/2026 Signature of the Lab Tutor: _____

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT
SCORE			

Lab # 4: Data Visualization and Text Analytics using Matplotlib and Word Cloud
Duration: 3 Hours

Objective:

The objective of this lab session is to introduce students to data visualization and basic text analytics using Matplotlib and Word Cloud.

Expected Outcomes: By the end of this lab session, students will be able to:

1. Matplotlib: Create and customize different types of plots, including line plots, bar charts, histograms, and scatter plots, to visualize and interpret data.
2. Word Cloud: Process text data, identify frequently occurring words, and create Word Cloud visualizations to explore patterns and major themes in text datasets.
3. Data Interpretation: Analyze visualizations and extract meaningful insights and patterns from text-based data.

Introduction to Matplotlib

Matplotlib is a Python library used to create **graphs and visualizations** from data.

In Big Data Analytics, visualization helps us:

- Identify trends
- Detect unusual values/outliers
- Compare categories
- Understand distributions
- Discover relationships between variables

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- Communicate analytical results

Basic import:

- `import matplotlib.pyplot as plt`

Matplotlib = the overall visualization library **pyplot** = a part/module of Matplotlib that provides the functions for making and controlling plots.

Basic Visualization

- 1. Line Plot:** A line plot is useful for showing trends over time.
- 2. Bar Chart:** Bar charts are useful for comparing different categories.
- 3. Histogram:** A histogram shows the distribution of numerical data.
- 4. Scatter Plot:** A scatter plot is useful for examining the relationship between two numerical variables.

Matplotlib with Pandas

This is the most important part for your Data science and analytics practical, because you have already learned Pandas in Lab 3.

<https://colab.research.google.com/drive/17F2LRGDfR6OCSrNxIAb642UEVvMthCOA?usp=sharing>

(WORK ON THIS NOTEBOOK)

Word Cloud in Data Science and Analytics

What is Word Cloud?

A Word Cloud is a visual representation of text data where words are displayed in different sizes.

- Frequently occurring words → appear larger
- Less frequent words → appear smaller
- It helps us quickly identify the most common or important words in a large collection of text.

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- For example, if a dataset contains many posts about Python:

PYTHON
DATA PYTHON
ANALYTICS
CODE DATA
PYTHON

- The word **Python** appears larger because it occurs more frequently.

Why is Word Cloud useful in Data Science and Analytics?

Big Data can contain a huge amount of **unstructured text**, such as:

- Reddit posts and comments
- Tweets/social media posts
- Customer reviews
- News articles
- Online discussions
- Survey responses
- Chat messages
- Feedback and complaints

Reading thousands or millions of text records manually is difficult.

A Word Cloud provides a **quick visual summary of frequently used words**.

Suppose we have thousands of customer reviews:

"The product is excellent..."

"Excellent quality and good service..."

"Good product but delivery was slow..."

"Excellent service..."

A Word Cloud might show:

EXCELLENT → very large

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GOOD → large

PRODUCT → medium

SERVICE → medium **DELIVERY**
→ small

This immediately gives us an idea about the **dominant words in the dataset**.

TASK:

DATA SCIENCE AND ANALYTICS
WORD CLOUD DATA

incident

"Phishing emails targeted university employees"

"Malware was detected on multiple computers"

"Suspicious network traffic was identified"

"Phishing attack compromised user credentials"

"Ransomware encrypted important company files"

"Unauthorized login attempts were detected"

"A malicious email contained a harmful attachment"

"Network traffic showed unusual activity"

"Several users reported suspicious emails"

"Malware infected computers through a downloaded file"

"An attacker attempted to steal login credentials"

"Phishing messages were sent to university students"

"A ransomware attack disrupted business operations"

"Security monitoring detected suspicious network connections"

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"Users received fake password reset emails"

"Malicious software was detected on the server"

"An employee clicked a phishing link"

"Unauthorized access was detected on a database"

"Attackers attempted to compromise user accounts"

"Suspicious files were discovered on multiple systems"

"Phishing emails requested sensitive information"

"Network monitoring detected unusual traffic patterns"

"A malware infection affected several workstations"

"An attacker used stolen credentials to access the network"

"Ransomware caused temporary loss of access to files"

"Security analysts detected a suspicious login"

"Employees received fraudulent emails containing links"

"Malicious code was found on a company server"

"An unauthorized user accessed sensitive data"

"Phishing attacks targeted employees through email"

"Suspicious activity was detected in the network"

"Attackers attempted to install malware on computers"

"A ransomware infection affected the organization"

"Several accounts were locked after failed login attempts"

"Sensitive information was exposed after unauthorized access"

"Security software detected malicious activity"

"Employees reported receiving suspicious messages"

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"An attacker attempted to steal confidential information"

"Malware was identified during a security scan"

"Network traffic indicated a possible cyber attack"

"Phishing links directed users to a fake website"

"Unauthorized access attempts targeted the database"

"Ransomware encrypted files on several computers"

"Security monitoring detected unusual user activity"

"Attackers sent malicious attachments through email"

"A compromised account was used to access company systems"

"Cybersecurity analysts investigated a suspected malware attack"

"Sensitive credentials were targeted by phishing emails"

"Suspicious network connections were blocked by security systems"

"An organization detected multiple attempted cyber attacks"

EXERCISE

Q1: Line Plot — Server Requests

A server records the number of requests it receives during different hours of the day.

hours = ["9 AM", "10 AM", "11 AM", "12 PM", "1 PM", "2 PM", "3 PM", "4 PM"]

requests = [450, 520, 610, 750, 680, 820, 900, 760]

Tasks:

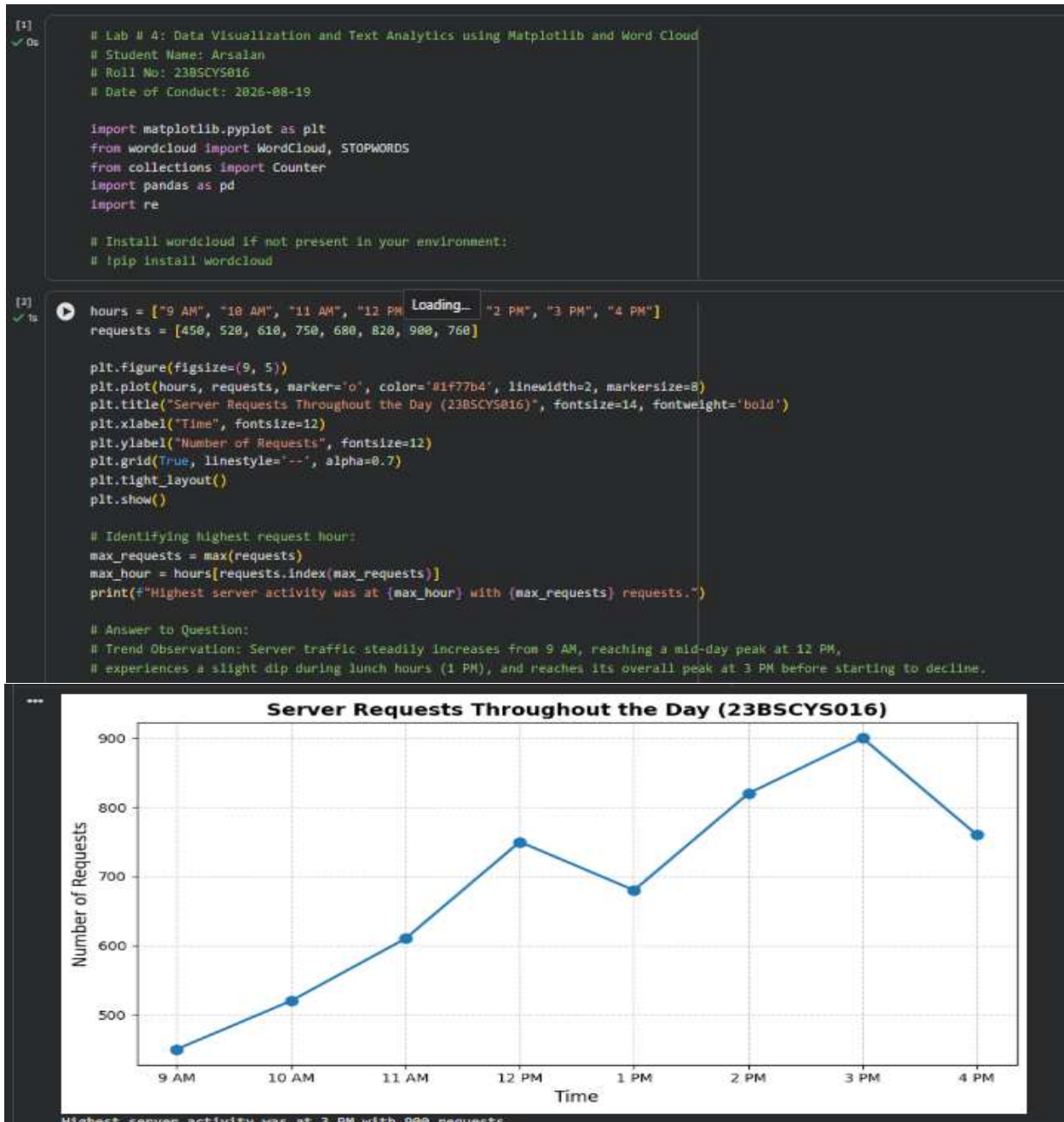
1. Create a line plot using Matplotlib.
2. Add a suitable title.
3. Label the X-axis as **Time**.
4. Label the Y-axis as **Number of Requests**.
5. Add markers to the data points.

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6. Add grid lines.
7. Identify the hour when the server received the highest number of requests.

Question:

What trend do you observe in the number of server requests throughout the day?



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Q 2: Bar Chart — Storage Usage by Department

A university stores data from different departments. The amount of storage used is given below:

departments = ["Cybersecurity", "Computer Science", "Electrical", "Software", "Telecom"]

storage = [850, 620, 450, 780, 530]

The storage values are in **GB**.

Tasks:

1. Create a bar chart.
2. Add suitable titles and axis labels.
3. Give the bars a suitable color.
4. Add grid lines.
5. Display the department names clearly.
6. Identify the department using the most storage.

Question:

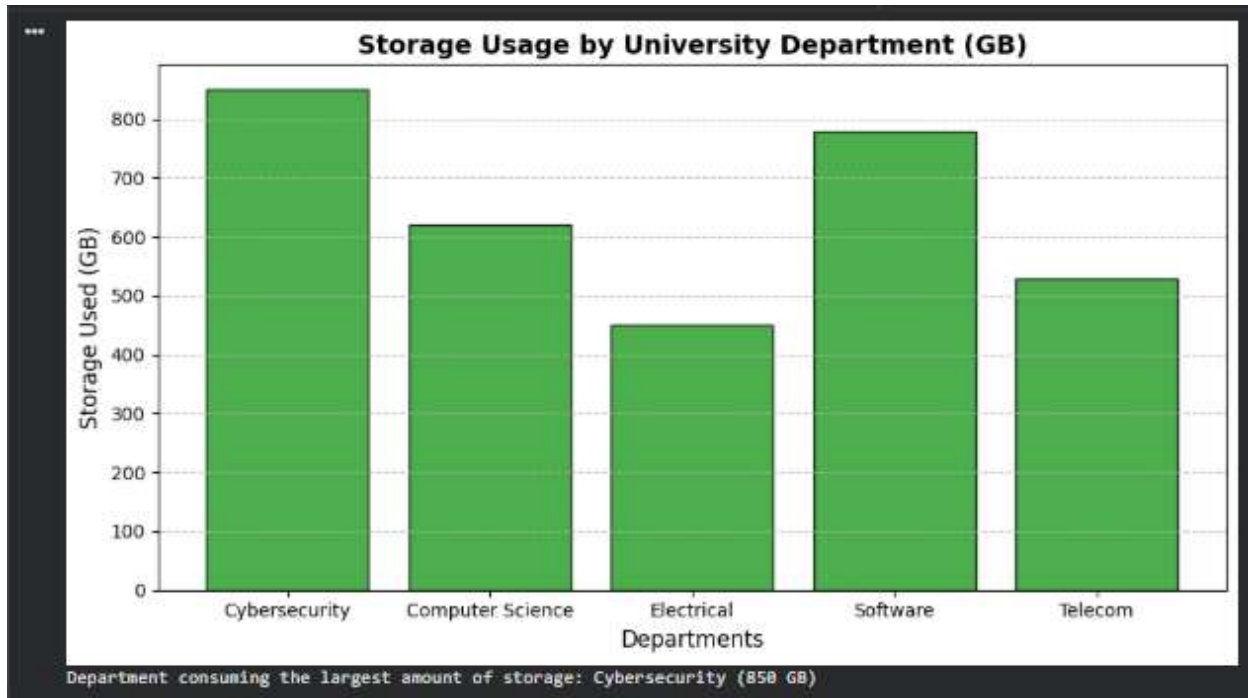
Which department consumes the largest amount of storage?

```
▶ departments = ["Cybersecurity", "Computer Science", "Electrical", "Software", "Telecom"]
  storage = [850, 620, 450, 780, 530]

plt.figure(figsize=(9, 5))
plt.bar(departments, storage, color='#2ca02c', edgecolor='black', alpha=0.85)
plt.title("Storage Usage by University Department (GB)", fontsize=14, fontweight='bold')
plt.xlabel("Departments", fontsize=12)
plt.ylabel("Storage Used (GB)", fontsize=12)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()

# Answer to Question:
max_dept = departments[storage.index(max(storage))]
print(f"Department consuming the largest amount of storage: {max_dept} ({max(storage)} GB)")
```

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Q3: Histogram — File Sizes

A data analyst records the sizes of files uploaded to a cloud storage system.

```
file_sizes = [  
    12, 15, 18, 20, 22, 25, 27, 30, 32, 35,  
    38, 40, 42, 45, 48, 50, 52, 55, 58, 60,  
    25, 30, 35, 40, 45, 50, 55, 60, 65, 70  
]
```

The values are in **MB**.

Tasks:

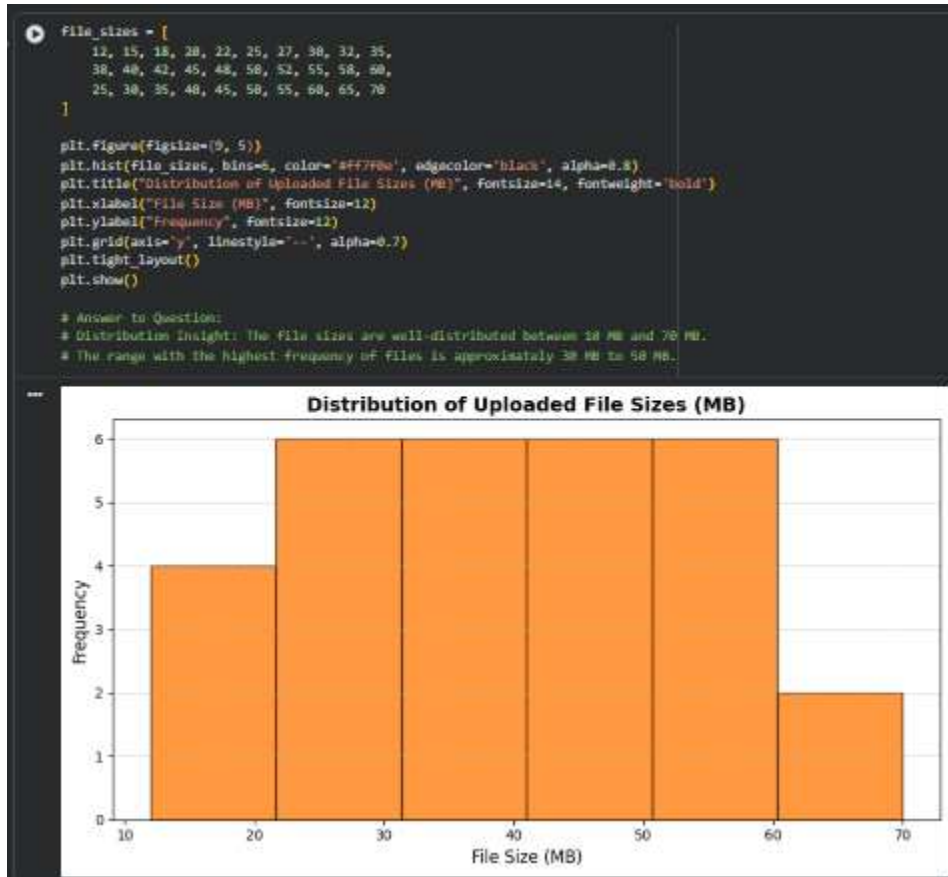
1. Create a histogram.
2. Use 6 bins.
3. Add a suitable title.
4. Label the X-axis as **File Size (MB)**.
5. Label the Y-axis as **Frequency**.

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6. Add an edge color to the bars.
7. Identify the file-size range containing the most files.

Question:

What does the histogram tell you about the distribution of file sizes



Q 4: Scatter Plot — Number of Users vs Data Generated

A cloud platform records the number of active users and the amount of data generated each day.

users = [100, 200, 300, 400, 500, 600, 700, 800, 900, 1000]

data_generated = [5, 11, 15, 22, 28, 31, 38, 45, 50, 57]

The data generated is measured in **GB**.

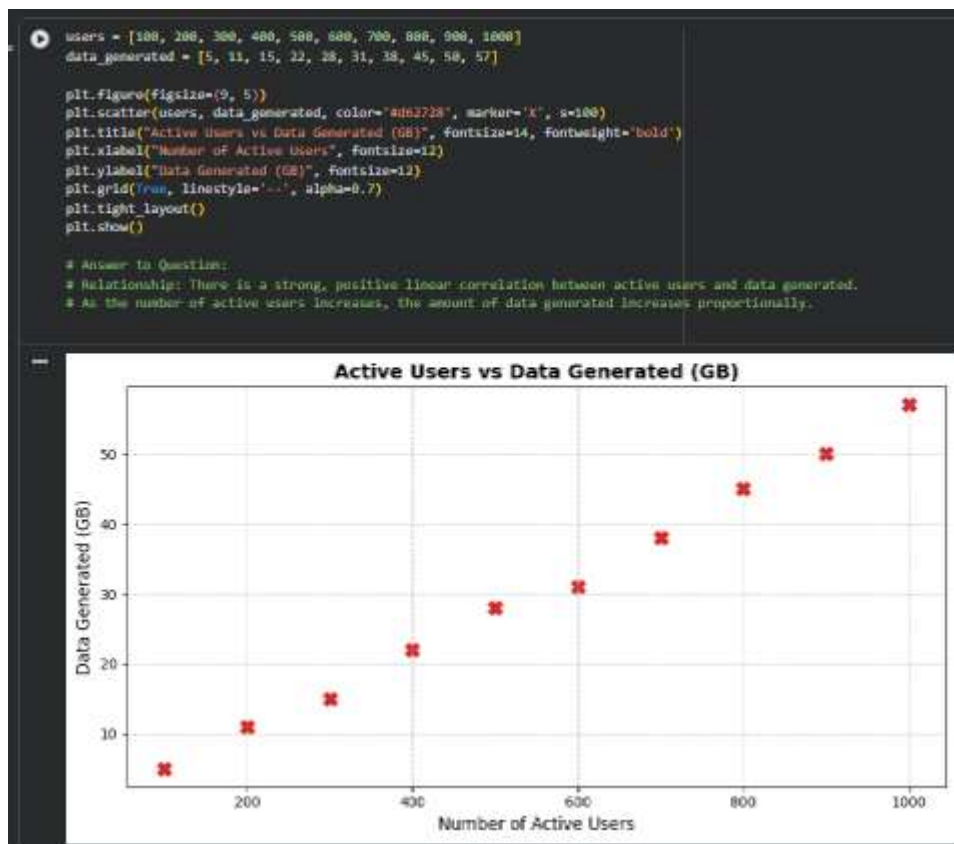
Tasks:

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1. Create a scatter plot.
2. Add a suitable title.
3. Label the X-axis as **Number of Users**.
4. Label the Y-axis as **Data Generated (GB)**.
5. Change the marker style.
6. Add grid lines.
7. Observe the relationship between the two variables.

Question:

As the number of users increases, what happens to the amount of data generated?



Q5: Choose the Correct Visualization

For each situation below, students must decide which visualization is most appropriate:

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Options:

Line Plot | Bar Chart | Histogram | Scatter Plot

1. Number of website visitors recorded every hour.
2. Number of attacks detected for different attack types.
3. Distribution of ages in a dataset of 1,000 users.
4. Relationship between CPU usage and processing time.
5. Monthly electricity consumption of a data center.
6. Number of files stored by different departments.

Task:

Write the appropriate visualization for each situation and give **one reason** for your choice.

• **Number of website visitors recorded every hour:**

- **Choice:** Line Plot
- **Reason:** Best suited for displaying trends and continuous data over time (temporal data).

• **Number of attacks detected for different attack types:**

- **Choice:** Bar Chart
- **Reason:** Ideal for comparing categorical discrete counts across different threat categories.

• **Distribution of ages in a dataset of 1,000 users:**

- **Choice:** Histogram
- **Reason:** Effective for showing frequency distributions of a continuous numerical variable split into range bins.

• **Relationship between CPU usage and processing time:**

- **Choice:** Scatter Plot
- **Reason:** Used to identify correlation and relationship patterns between two numeric variables.

• **Monthly electricity consumption of a data center:**

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- **Choice:** Line Plot
- **Reason:** Displays longitudinal sequential time-series patterns across consecutive months.
- **Number of files stored by different departments:**
 - **Choice:** Bar Chart
 - **Reason:** Best for distinct categorical dynamic metric comparisons.

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Q6: Word Frequency Analysis Using

the same cybersecurity dataset:

Tasks:

1. Convert all text to lowercase.
2. Split the text into individual words.
3. Count the frequency of each word.
4. Display the **10 most frequently occurring words**.
5. Create a bar chart showing the top 10 words.

```
# Raw text dataset from lab exercise
Incidents = [
    "Phishing emails targeted university employees", "Malware was detected on multiple computers",
    "Suspicious network traffic was identified", "Phishing attack compromised user credentials",
    "Ransomware encrypted important company files", "Unauthorized login attempts were detected",
    "A malicious email contained a harmful attachment", "Network traffic showed unusual activity",
    "Several users reported suspicious emails", "Malware infected computers through a downloaded file",
    "An attacker attempted to steal login credentials", "Phishing messages were sent to university students",
    "A ransomware attack disrupted business operations", "Security monitoring detected suspicious network connections",
    "Users received fake password reset emails", "Malicious software was detected on the server",
    "An employee clicked a phishing link", "Unauthorized access was detected on a database",
    "Attackers attempted to compromise user accounts", "Suspicious files were discovered on multiple systems",
    "Phishing emails requested sensitive information", "Network monitoring detected unusual traffic patterns",
    "A malware infection affected several workstations", "An attacker used stolen credentials to access the network",
    "Ransomware caused temporary loss of access to files", "Security analysts detected a suspicious login",
    "Employees received fraudulent emails containing links", "Malicious code was found on a company server",
    "An unauthorized user accessed sensitive data", "Phishing attacks targeted employees through email",
    "Suspicious activity was detected in the network", "Attackers attempted to install malware on computers",
    "A ransomware infection affected the organization", "Several accounts were locked after failed login attempts",
    "Sensitive information was exposed after unauthorized access", "Security software detected malicious activity",
    "Employees reported receiving suspicious messages", "An attacker attempted to steal confidential information",
    "Malware was identified during a security scan", "Network traffic indicated a possible cyber attack",
    "Phishing links directed users to a fake website", "Unauthorized access attempts targeted the database",
    "Ransomware encrypted files on several computers", "Security monitoring detected unusual user activity",
    "Attackers sent malicious attachments through email", "A compromised account was used to access company systems",
    "Cybersecurity analysts investigated a suspected malware attack", "Sensitive credentials were targeted by phishing emails",
    "Suspicious network connections were blocked by security systems", "An organization detected multiple attempted cyber attacks"
]

# 1 & 2: lowercase and tokenization into words (filtering common stopwords)
all_text = " ".join(Incidents).lower()
words = re.findall(r'\b[a-z]{1,}\b', all_text)

# Standard English Stopwords filtering
stop_words = set(STOPWORDS).union({'was', 'were', 'been', 'through', 'during', 'used', 'using', 'several'})
filtered_words = [word for word in words if word not in stop_words]

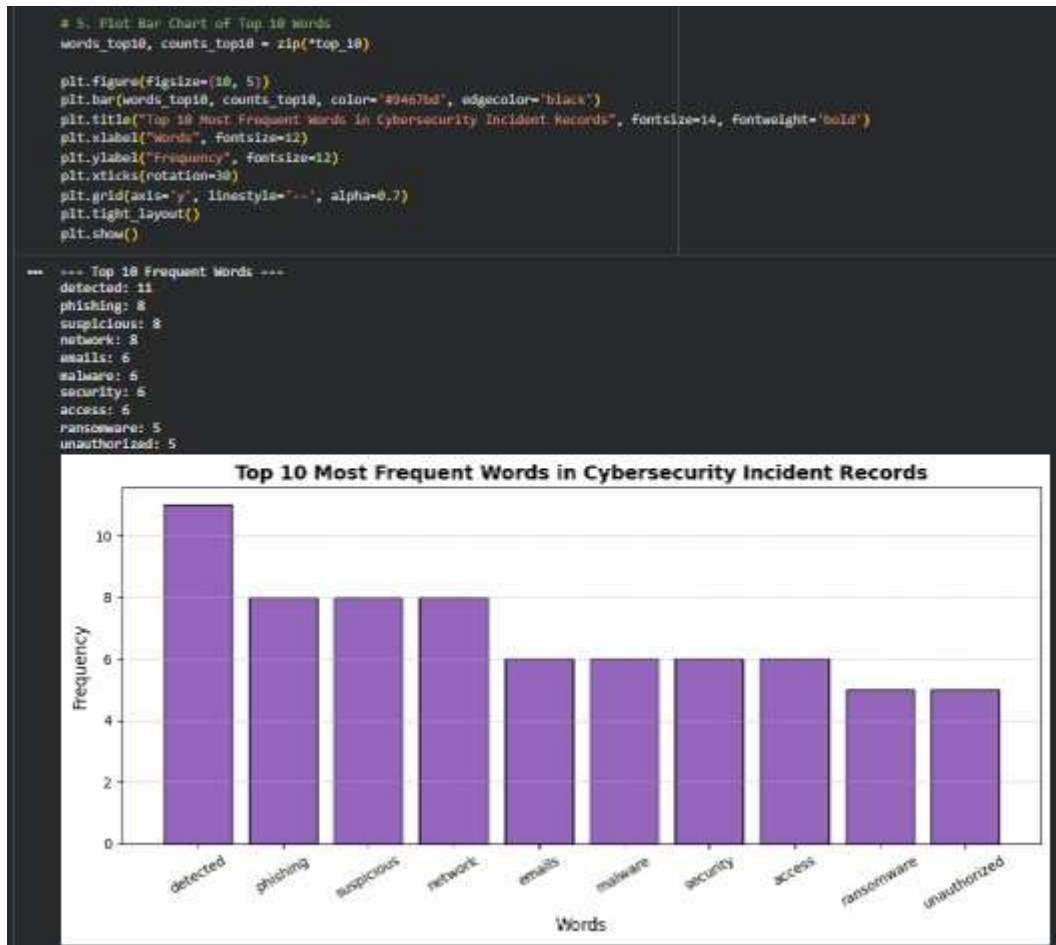
# 3: Count frequencies
word_counts = Counter(filtered_words)

# 4: Display 10 most frequent words
top_10 = word_counts.most_common(10)
print("--- Top 10 frequent words ---")
for word, count in top_10:
    print(f"{word}: {count}")

# 5: Plot Bar Chart of Top 10 words
words_top10, counts_top10 = zip(*top_10)

plt.figure(figsize=(10, 5))
plt.bar(words_top10, counts_top10, color='#9467bd', edgecolor='black')
plt.title("Top 10 Most Frequent words in Cybersecurity Incident Records", fontsize=14, fontweight='bold')
plt.xlabel("words", fontsize=12)
```

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Q7: Open-Ended Data Science and Analytics Task

You are working as a **data analyst in a cybersecurity organization**. You have been given a large collection of cybersecurity incident descriptions.



Microsoft Excel Worksheet
(code)

Perform a basic text analytics process using Python.

Your solution must include:

- Loading the dataset using Pandas
- Exploring the dataset
- Finding the top 10 words

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- Creating a Word Cloud
- Creating a bar chart of word frequencies
- Identifying at least **three important observations**

Based on your analysis, what are the major cybersecurity issues represented in the dataset?

```
# Loading raw data into a Pandas DataFrame
df_incidents = pd.DataFrame({'Incident': incidents})

# Generate word frequency text data from DataFrame
corpus = " ".join(df_incidents['Incident']).lower()

# Word Cloud Customization: Dark background with vibrant security-themed colormap ('plasma')
wc = WordCloud(
    width=800,
    height=500,
    background_color='black',      # Custom Background
    colormap='plasma',             # Custom Color Palette
    max_words=80,
    stopwords=stop_words,
    random_state=42
).generate(corpus)

plt.figure(figsize=(12, 6))
plt.imshow(wc, interpolation='bilinear')
plt.axis('off')
plt.title("Cybersecurity Incident Word Cloud - Student: Arsalan (2985V5816)", fontsize=15, fontweight='bold', pad=15)
plt.tight_layout()
plt.show()

# Key Observations:
print("\n--- Key Incident Analysis Observations ---")
print("1. Social Engineering Dominance: 'Phishing' and 'Emails' are among the highest frequency terms, indicating social engineering is a primary attack vector.")
print("2. System Exposure & Malware: 'Malware', 'Ransomware', and 'Detected' highlight constant endpoint exposure and malicious software deployments.")
print("3. Unauthorized Credential Abuse: Frequent occurrences of 'Unauthorized', 'Access', 'Stolen', and 'Credentials' indicate recurring identity and authentication breaches.")
```

